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DETECTION DEVICE

Abstract

According to an aspect, a detection device includes: an optical sensor comprising photodetection elements arranged in a planar configuration; a light source configured to emit light to the photodetection elements; and an object placement member. The light source, the object placement member, and the optical sensor are arranged in the order as listed. The optical sensor is configured to obtain baseline data in an initial state after power-on. The detection device is configured to: obtain difference data based on a difference between the baseline data and sensor values obtained from the photodetection elements at intervals of a predetermined period of time; and output, when the difference data has exceeded a first threshold, either an alert indicating detection of the objects to be detected in an initial moisture region indicating presence of moisture in the initial state, or an alert indicating detection of the objects to be detected.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of priority from Japanese Patent Application No. 2024-031590 filed on Mar. 1, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Technical Field

[0002] What is disclosed herein relates to a detection device.

2. Description of the Related Art

[0003] Japanese Patent Application Laid-open Publication No. 2018-033430 discloses an image acquisition device that includes an optical sensor, a container to contain microorganisms and a culture medium, and a light source, and acquires, over time, images indicating a growth of the microorganisms in the container.

[0004] In such a detection device, if moisture (e.g., beads of water) adheres to the container, an object to be detected as a detection target may be difficult to be detected well.

[0005] For the foregoing reasons, there is a need for a detection device capable of improving the accuracy of the detection.

SUMMARY

[0006] According to an aspect, a detection device includes: an optical sensor comprising a plurality of photodetection elements arranged in a planar configuration; a light source configured to emit light to the photodetection elements; and an object placement member that has a light-transmitting property and on which a plurality of objects to be detected are to be placed. The light source, the object placement member, and the optical sensor are arranged in the order as listed. The optical sensor is configured to obtain baseline data in an initial state after power-on. The detection device is configured to: obtain difference data based on a difference between the baseline data and sensor values obtained from the photodetection elements at intervals of a predetermined period of time; and output, when the difference data has exceeded a first threshold, either an alert indicating detection of the objects to be detected in an initial moisture region indicating presence of moisture in the initial state, or an alert indicating detection of the objects to be detected.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a sectional view schematically illustrating a detection device according to an embodiment;

[0008] FIG. 2 is a block diagram illustrating a configuration example of the detection device according to the embodiment;

[0009] FIG. 3 is a circuit diagram illustrating a sensor pixel;

[0010] FIG. 4 is a block diagram illustrating a configuration example of a detection circuit according to the embodiment;

[0011] FIG. 5 is a block diagram illustrating a configuration example of a sensor control circuit according to the embodiment;

[0012] FIG. 6 is a schematic diagram schematically illustrating states of growth of objects to be detected;

[0013] FIG. 7 is a graph schematically illustrating a relation between a sensor value and time in detection of the objects to be detected;

[0014] FIG. 8 is a graph schematically illustrating a relation between difference data and time in the detection of the objects to be detected;

[0015] FIG. 9 is an explanatory chart for explaining the sensor value, the difference data, and the detected objects in the detection of the objects to be detected;

[0016] FIG. 10 is a schematic diagram schematically illustrating states of evaporation of moisture;

[0017] FIG. 11 is a graph schematically illustrating a relation between the sensor value and time in detection of the moisture;

[0018] FIG. 12 is a graph schematically illustrating a relation between the difference data and time in the detection of the moisture;

[0019] FIG. 13 is an explanatory chart for explaining the sensor values, the difference data, and a detected initial moisture region in the detection of the moisture;

[0020] FIG. 14 is a graph schematically illustrating a relation between the sensor value and time in the detection of the moisture according to a modification of the embodiment;

[0021] FIG. 15 is a graph schematically illustrating a relation between the difference data and time in the detection of the moisture according to the modification;

[0022] FIG. 16 is an explanatory chart for explaining the sensor values, the difference data, and the detected initial moisture region in the detection of the moisture according to the modification;

[0023] FIG. 17 is a graph schematically illustrating a relation between the sensor value and time in the detection of the objects to be detected and the moisture;

[0024] FIG. 18 is a graph schematically illustrating a relation between the difference data and time in the detection of the objects to be detected and the moisture;

[0025] FIG. 19 is a schematic diagram schematically illustrating the states of the growth of the objects to be detected and the states of the evaporation of the moisture;

[0026] FIG. 20 is an explanatory chart for explaining the sensor values, the difference data, and the objects to be detected and the initial moisture region that have been detected in the detection of the objects to be detected and the moisture;

[0027] FIG. 21 is a flowchart for explaining a method for obtaining baseline data;

[0028] FIG. 22 is a flowchart for explaining a method for obtaining the difference data;

[0029] FIG. 23 is a flowchart for explaining a method for detecting the objects to be detected and the output of alerts; and

[0030] FIG. 24 is a flowchart for explaining a method for detecting the moisture.

DETAILED DESCRIPTION

[0031] The following describes a mode (embodiment) for carrying out the present disclosure in detail with reference to the drawings. The present disclosure is not limited to the description of the embodiment given below. Components described below include those easily conceivable by those skilled in the art or those substantially identical thereto. In addition, the components described below can be combined as appropriate. What is disclosed herein is merely an example, and the present disclosure naturally encompasses appropriate modifications easily conceivable by those skilled in the art while maintaining the gist of the present disclosure. To further clarify the description, the drawings may schematically illustrate, for example, widths, thicknesses, and shapes of various parts as compared with actual aspects thereof. However, they are merely examples, and interpretation of the present disclosure is not limited thereto. The same component as that described with reference to an already mentioned drawing is denoted by the same reference numeral through the present disclosure and the drawings, and detailed description thereof may not be repeated where appropriate.

[0032] In the present disclosure, in expressing an aspect of disposing another structure on or above a certain structure, a case of simply expressing “on” includes both a case of disposing the other structure immediately on the certain structure so as to contact the certain structure and a case of

disposing the other structure above the certain structure with still another structure interposed therebetween, unless otherwise specified.

Embodiment

[0033] FIG. 1 is a sectional view schematically illustrating a detection device according to an embodiment. As illustrated in FIG. 1, a detection device 1 includes an optical sensor 10, an optical filter layer 50, a container 110 for accommodating an object to be detected 100, and a light source 80. In the detection device 1, the light source 80, the container 110 (object to be detected 100), the optical filter layer 50, and the optical sensor 10 are arranged in this order.

[0034] The object to be detected 100 is, for example, micro-objects such as bacteria. The bacteria or the like that have been cultured on a culture medium 102 (e.g., agar) and grown into a clump large enough to be visible may be referred to as a colony. The detection device 1 is a biosensor that detects the micro-objects such as the bacteria. The object to be detected 100 is not limited to the bacteria and may be other micro-objects such as cells.

[0035] The container 110 includes a container body 111 and a cover member 112. The container 110 is a Petri dish, for example. The container 110 is light-transmitting. The container body 111 contains the culture medium 102, and the object to be detected 100 is cultured on the culture medium 102. That is, the container 110 (at least the container body 111, of the container body 111 and the cover member 112) is an object placement member that has a light-transmitting property and in which (on which) a plurality of the objects to be detected 100 can be placed.

[0036] In the present embodiment, the container 110 is placed upside down with respect to a normal container. That is, the normal container is placed such that the container body is located on the lower side and the cover member is located on the upper side. In contrast, the container 110 according to the present embodiment is placed such that the container body 111 is located on the upper side and the cover member 112 is located on the lower side. Specifically, the objects to be detected 100 such as the bacteria are placed on the upper side of the culture medium 102 and cultured, and when imaging the objects to be detected 100, the container 110 is placed upside down such that the objects to be detected 100 is placed on the lower side of the culture medium 102. The objects to be detected 100 serving as detection targets and the culture medium 102 are contained in the container 110 and placed between the optical sensor 10 and the light source 80.

[0037] In the container 110, water vapor may be generated from the culture medium 102, and moisture 103 (e.g., beads of water) may adhere to the culture medium 102 or the cover member 112 of the container 110. Alternatively, the moisture 103 may adhere to the outer surface of the container 110 (upper surface of the container body 111 in FIG. 1). In this case, the intensity of light from the light source 80 changes when the light passes through the moisture 103, which may make it difficult to distinguish between the moisture 103 and the objects to be detected 100 by using sensor values So detected by the optical sensor 10. Alternatively, if the objects to be detected 100 are placed so as to overlap the moisture 103, it may be difficult to detect well the objects to be detected 100. A detailed method for detecting the objects to be detected 100 and the moisture 103 will be described later with reference to FIG. 6 and the subsequent drawings.

[0038] The optical sensor 10 is a detection device that includes a plurality of photodetection elements 30 arranged in a plane. The photodetection elements 30 are photodiodes, for example. More specifically, the photodetection elements 30 are positive-intrinsic-negative (PIN) photodiodes using inorganic semiconductors or organic photodiodes (OPDs) using organic semiconductors.

[0039] The optical filter layer 50 is a light directivity control element disposed between a plurality of light-emitting elements 82 (light source 80) and the photodetection elements 30 (optical sensor 10). More specifically, the optical filter layer 50 is provided between the photodetection elements 30 of the optical sensor 10 and the container 110. The optical filter layer 50 is disposed so as to face the photodetection elements 30 of the optical sensor 10. The optical filter layer 50 is an optical element that transmits, toward the photodetection elements 30, components of light emitted from the light-emitting elements 82 and traveling in a direction orthogonal to the optical sensor 10. The

optical filter layer **50** is also called collimating apertures or a collimator.

[0040] The light source **80** includes a light source substrate **81** and the light-emitting elements **82**. The light-emitting elements **82** are point light sources provided correspondingly to the photodetection elements **30** of the optical sensor **10**. The light-emitting elements **82** are provided on the light source substrate **81** and arranged so as to face the photodetection elements **30** of the optical sensor **10**. Each of the light-emitting elements **82** is configured as a light-emitting diode (LED), for example.

[0041] The light emitted from the light-emitting elements **82** passes through the cover member **112**, the culture medium **102**, the container body **111**, and the optical filter layer **50**, and is emitted toward the photodetection elements **30** of the optical sensor **10**. The intensity of the light applied to the photodetection elements **30** differs between an area overlapping the objects to be detected **100** and an area not overlapping the objects to be detected **100**. As a result, the optical sensor **10** can image the objects to be detected **100**. Alternatively, when the moisture **103** is present, the intensity of the light applied to the photodetection elements **30** differs between an area overlapping the moisture **103** and an area not overlapping the moisture **103**. As a result, the optical sensor **10** can detect the presence or absence of the moisture **103**.

[0042] FIG. **2** is a block diagram illustrating a configuration example of the detection device according to the embodiment. As illustrated in FIG. **2**, the detection device **1** further includes a control circuit **70** that controls the optical sensor **10** and the light source **80**. The control circuit **70** synchronously (or non-synchronously) controls an operation of detecting the objects to be detected **100** by the optical sensor **10** and an operation of lighting the light-emitting elements **82** by the light source **80**. The control circuit **70** includes, for example, a micro-controller unit (MCU), a random-access memory (RAM), an electrically erasable programmable read-only memory (EEPROM), and a read-only memory (ROM).

[0043] The optical sensor **10** includes an array substrate **2**, a plurality of sensor pixels **3** (photodetection elements **30**) formed on the array substrate **2**, gate line drive circuits **15A** and **15B**, a signal line drive circuit **16A**, and a detection circuit **11**.

[0044] The array substrate **2** is formed using a substrate **21** as a base. Each of the sensor pixels **3** is configured with a corresponding one of the photodetection elements **30**, a plurality of transistors, and various types of wiring. The array substrate **2** with the photodetection elements **30** formed thereon is a drive circuit board for driving the sensor for each predetermined detection area and is also called a backplane or an active-matrix substrate.

[0045] The substrate **21** has a detection area **AA** and a peripheral area **GA**. The sensor pixels **3** (photodetection elements **30**) are arranged in a matrix having a row-column configuration in the detection area **AA**. The gate line drive circuits **15A** and **15B**, the signal line drive circuit **16A**, and the detection circuit **11** are provided in the peripheral area **GA**.

[0046] The detection circuit **11** is a circuit that supplies control signals **Sa**, **Sb**, and **Sc** to the gate line drive circuits **15A** and **15B**, and the signal line drive circuit **16A**, respectively, to control operations of these circuits. Specifically, the gate line drive circuits **15A** and **15B** output gate drive signals to gate lines **GLS** (refer to FIG. **3**) based on the control signals **Sa** and **Sb**. The signal line drive circuit **16A** electrically couples a signal line **SLS** (refer to FIG. **3**) selected based on the control signal **Sc** to the detection circuit **11**. The detection circuit **11** includes a signal processing circuit that performs signal processing on a detection signal **Vdet** from each of the photodetection elements **30**. The detection circuit **11** includes a readout integrated circuit (ROIC).

[0047] The photodetection elements **30** included in the sensor pixels **3** perform detection in response to the gate drive signals supplied from the gate line drive circuits **15A** and **15B**. Each of the photodetection elements **30** outputs an electrical signal corresponding to the light emitted thereto as the detection signal **Vdet** to the signal line drive circuit **16A**. The detection circuit **11** is electrically coupled to the photodetection elements **30** via the signal line drive circuit **16A**. The detection circuit **11** performs the signal processing on the detection signals **Vdet** from the

photodetection elements **30** and outputs the sensor values S_o based on the detection signals V_{det} to the control circuit **70**. Thus, the detection device **1** detects information on the objects to be detected **100**.

[0048] FIG. **3** is a circuit diagram illustrating the sensor pixel. As illustrated in FIG. **3**, the sensor pixel **3** includes the photodetection element **30**, a capacitive element C_a , and a transistor TrS . The transistor TrS is provided correspondingly to the photodetection element **30**. The transistor TrS is configured as a thin-film transistor, and in this example, configured as an n-channel metal oxide semiconductor (MOS) thin-film transistor (TFT). The gate of the transistor TrS is coupled to the gate line GLS . The source of the transistor TrS is coupled to the signal line SLS . The drain of the transistor TrS is coupled to the anode of the photodetection element **30** and the capacitive element C_a .

[0049] The cathode of the photodetection element **30** is supplied with a power supply potential SVS from the detection circuit **11**. The capacitive element C_a is supplied with a reference potential $VR1$ serving as an initial potential of the capacitive element C_a from the detection circuit **11**.

[0050] When the sensor pixel **3** is irradiated with light, a current corresponding to the intensity of the light flows through the photodetection element **30**. As a result, an electric charge is stored in the capacitive element C_a .

[0051] Turning on the transistor TrS causes a current corresponding to the electric charge stored in the capacitive element C_a to flow through the signal line SLS . The signal line SLS is coupled to the detection circuit **11** via the signal line drive circuit **16A**. Thus, the optical sensor **10** of the detection device **1** can detect a signal corresponding to the intensity of the light received by the photodetection element **30** for each of the sensor pixels **3**.

[0052] The transistor TrS is not limited to the n-type TFT and may be configured as a p-type TFT. The pixel circuit of the sensor pixel **3** illustrated in FIG. **3** is merely exemplary. The sensor pixel **3** may be provided with a plurality of transistors corresponding to one photodetection element **30**.

[0053] FIG. **4** is a block diagram illustrating a configuration example of the detection circuit according to the embodiment. As illustrated in FIG. **4**, the detection circuit **11** includes a detection signal amplitude adjustment circuit **41**, an analog-to-digital (A/D) conversion circuit **42**, a signal processing circuit **43**, and a detection timing control circuit **44**. In the detection circuit **11**, the detection timing control circuit **44** performs control to synchronously operate the detection signal amplitude adjustment circuit **41**, the A/D conversion circuit **42**, and the signal processing circuit **43** based on a control signal supplied from the control circuit **70** (refer to FIG. **2**).

[0054] The detection signal amplitude adjustment circuit **41** is a circuit that adjusts the amplitude of the detection signal V_{det} output from the photodetection element **30** and is configured with an amplifier, for example. The A/D conversion circuit **42** converts analog signals output from the detection signal amplitude adjustment circuit **41** into digital signals. The signal processing circuit **43** performs signal processing on the digital signals from the A/D conversion circuit **42** and transmits the sensor values S_o to the control circuit **70**.

[0055] Referring back to FIG. **2**, the light source **80** includes a light source drive circuit **12** that drives the light-emitting elements **82** mounted on the light source substrate **81**. The light-emitting elements **82** are arranged in a matrix having a row-column configuration in an area of the light source substrate **81** overlapping the detection area AA . The light source drive circuit **12** supplies a power supply voltage (an anode power supply potential and a cathode power supply potential) to the light-emitting elements **82** based on a control signal S_d from the control circuit **70** (light source control circuit **72**). This operation switches the light-emitting elements **82** between on (lit state) and off (unlit state).

[0056] The number and arrangement of the light-emitting elements **82** can be changed as appropriate. The light-emitting elements **82** may emit light in a single color or may be configured to emit light having several different wavelengths. The lighting pattern of the light-emitting elements **82** can also be changed as appropriate depending on the state of the objects to be detected

100 serving as the detection target. The light-emitting elements **82** may be simultaneously turned on or may be turned on in a time-division manner on a predetermined area basis.

[0057] The control circuit **70** includes a sensor control circuit **71** that controls the optical sensor **10**, the light source control circuit **72** that controls the light source **80**, and a communication circuit **73**. The sensor control circuit **71** and the light source control circuit **72** control the optical sensor **10** and the light source **80**, respectively, so that the detection operation of the optical sensor **10** and the lighting operation of the light source **80** are synchronously performed.

[0058] The communication circuit **73** couples the control circuit **70** to an external circuit **85** in a wired or wireless manner. The external circuit **85** is a personal computer (PC), for example. The external circuit **85** is not limited thereto and may be a portable device such as a tablet computer or a smartphone. Thus, information on the objects to be detected **100** or the moisture **103** detected by the detection device **1** is output to the external circuit **85** via the communication circuit **73**. By operating the external circuit **85**, a user enters various conditions to be used for the detection, such as a start time and an end time of the detection of the detection device **1**, and/or a first threshold Th_c and a second threshold Th_w .

[0059] FIG. 5 is a block diagram illustrating a configuration example of the sensor control circuit according to the embodiment. As illustrated in FIG. 5, the sensor control circuit **71** includes an arithmetic circuit **74**, an image generation circuit **75**, a determination circuit **76**, an alert output circuit **77**, and a storage circuit **78**.

[0060] The arithmetic circuit **74** calculates difference data $Diff$ based on baseline data (sensor values $So(T0)$) and data of the sensor values $So(Ti)$ received from the optical sensor **10**. The baseline data is data of the sensor values So obtained by scanning the photodetection elements **30** of the optical sensor **10** in an initial state (for example, at time $T0$ after power-on). The difference data $Diff$ will be described later with reference to FIGS. 8 and 9.

[0061] The image generation circuit **75** generates an image based on the difference data $Diff$ or the data of the sensor values So .

[0062] The determination circuit **76** determines the presence or absence of the objects to be detected **100** or the presence or absence of the moisture **103** based on the image data of the image generation circuit **75** and the first threshold Th_c and the second threshold Th_w stored in advance in the storage circuit **78**. The determination circuit **76** also determines regions, such as an initial moisture region $R1$ indicating the presence of the moisture **103** in the initial state, and a moisture evaporation region $R2$ indicating that the moisture **103** has evaporated, and the positional relation therebetween.

[0063] The alert output circuit **77** outputs an alert indicating the detection of the objects to be detected **100** when the objects to be detected **100** are detected. Alternatively, the alert output circuit **77** outputs an alert indicating the positional relation between the objects to be detected **100** and the moisture **103**. A display device **86** of the external circuit **85** receives the alert from the alert output circuit **77** and displays the image indicating the detected position of the objects to be detected **100**.

[0064] The storage circuit **78** stores therein the sensor values So and the baseline data detected by the optical sensor **10**, the difference data $Diff$ calculated by the arithmetic circuit **74**, and the various preset conditions such as the first threshold Th_c and the second threshold Th_w .

[0065] The sensor control circuit **71** illustrated in FIG. 5 is illustrated schematically for ease of understanding. The circuits included in the sensor control circuit **71** are not limited to being configured such that each of the circuits is provided as an individual circuit and may be configured as one integrated circuit (IC).

[0066] The following describes a detection method of the detection device **1**. As described above, the detection device **1** detects the objects to be detected **100** cultured on the culture medium **102**. However, if the moisture **103** adheres to the container **110** or the culture medium **102**, the detection device **1** may detect the moisture **103**. There is a demand to reduce false detection of the moisture **103** in the detection of the objects to be detected **100** cultured on the culture medium **102**.

Alternatively, there is a demand to allow the user to visually check the image when the objects to be detected **100** are located so as to overlap the moisture **103**.

[0067] The following sequentially describes a method for detecting the objects to be detected **100** when the moisture **103** is not present, a method for detecting the moisture **103** when the objects to be detected **100** are not present (or not fully grown), and a method for detecting the objects to be detected **100** when they are located so as to overlap the moisture **103**.

[0068] FIG. **6** is a schematic diagram schematically illustrating states of growth of the objects to be detected. FIG. **7** is a graph schematically illustrating a relation between the sensor value and time in the detection of the objects to be detected. FIG. **8** is a graph schematically illustrating a relation between the difference data and time in the detection of the objects to be detected. FIG. **9** is an explanatory chart for explaining the sensor value, the difference data, and the detected objects in the detection of the objects to be detected.

[0069] In FIG. **6**, the intensity of the light transmitted through the container **110** and the culture medium **102** and received by the optical sensor **10** (hereinafter, referred to as “transmitted light intensity”) is schematically illustrated in different gray shades. That is, the degree of the gray shade illustrated in FIG. **6** corresponds to the magnitude of the sensor values S_o detected by the optical sensor **10**. The shade is illustrated closer to white as the transmitted light intensity (sensor value S_o) is higher, and illustrated closer to black as the transmitted light intensity (sensor value S_o) is lower. FIGS. **7** and **8** illustrate the sensor value S_o and the difference data $Diff$ in the area overlapping the objects to be detected **100** in FIG. **6**.

[0070] As illustrated in FIGS. **6** and **7**, the detection device **1** scans the photodetection elements **30** of the optical sensor **10** at predetermined intervals of time to detect changes over time of the objects to be detected **100** cultured on the culture medium **102**. FIG. **6** schematically illustrates the detection results of the objects to be detected **100** at three times T_0 , T_1 , and T_i . However, the detection device **1** is not limited to detecting the objects to be detected **100** three times, and repeatedly detects the objects to be detected **100** at the predetermined intervals of time (for example, at intervals of substantially 5 to 10 minutes).

[0071] As illustrated in FIG. **6**, at time T_0 in the initial state, the objects to be detected **100** have not grown, so that the transmitted light intensity across the culture medium **102** in the container **110** is substantially constant. In the following description, the sensor values S_o at the initial state (time T_0) are represented as the baseline data (sensor values $S_o(T_0)$). At times T_1 and T_i when predetermined times have elapsed from time T_0 , the objects to be detected **100** have grown, and the transmitted light intensity in the area overlapping the objects to be detected **100** gradually decreases. To facilitate understanding of the description, the description will be made assuming that the transmitted light intensity across the culture medium **102** is constant within the container **110** and does not change (or changes at a negligibly small degree) over time.

[0072] As illustrated in FIG. **7**, the sensor value S_o exhibits a substantially constant value because the objects to be detected **100** have not grown for a predetermined period of time starting from time T_0 . As the objects to be detected **100** grow, the sensor value S_o becomes smaller as the transmitted light intensity decreases.

[0073] As illustrated in FIG. **8**, the difference data $Diff$ exhibits a tendency of change inverted from that of the sensor value S_o illustrated in FIG. **7**. That is, the difference data $Diff$ exhibits a substantially constant value ($Diff=0$) during the period in which the objects to be detected **100** have not grown for the predetermined period of time starting from time T_0 . As the objects to be detected **100** grow, the difference data $Diff$ becomes larger as the sensor value S_o decreases.

[0074] As illustrated in FIGS. **8** and **9**, the arithmetic circuit **74** (refer to FIG. **5**) calculates the difference data $Diff$ based on the baseline data (sensor values $S_o(T_0)$) received from the optical sensor **10** and the sensor values $S_o(T_i)$ at time T_i . The difference data $Diff$ is expressed as $Diff=S_o(T_0)-S_o(T_i)$. In FIG. **7**, the baseline data is the sensor values S_o at time 0 h (T_0).

[0075] In FIG. **9**, the shade is illustrated closer to black as the difference data $Diff$ is smaller and

illustrated closer to white as the difference data Diff is larger. The area illustrated in black in FIG. 9 indicates that the difference data Diff is substantially zero.

[0076] As illustrated in FIG. 9, the difference data Diff is larger (illustrated in white) in the area overlapping the objects to be detected **100**. The difference data Diff is smaller (illustrated in black) in the area (culture medium **102**) not overlapping the objects to be detected **100**.

[0077] The determination circuit **76** (refer to FIG. 5) compares the difference data Diff with the first threshold Th_c stored in advance in the storage circuit **78**. The first threshold Th_c is a reference value indicating that the objects to be detected **100** have been detected. In the area overlapping the objects to be detected **100**, the difference data Diff is larger than the first threshold Th_c (Th_c<Diff). In the area not overlapping the objects to be detected **100**, the difference data Diff is smaller than the first threshold Th_c (Diff<Th_c). The determination circuit **76** determines that the objects to be detected **100** have been detected when the difference data Diff exceeds the first threshold Th_c.

[0078] In the example illustrated in FIGS. 6 to 9, the moisture **103** is not present at time T_i, and an area (initial moisture region R1 to be described later) indicating the presence of the moisture **103** in the initial state (at time T₀) is not detected. Therefore, the determination circuit **76** determines that the moisture **103** is not present and the objects to be detected **100** have been detected. The alert output circuit **77** outputs the alert indicating the detection of the objects to be detected **100**. The display device **86** of the external circuit **85** displays the image indicating the detected position of the objects to be detected **100** along with the alert indicating the detection of the objects to be detected **100**.

[0079] The following describes the method for detecting the moisture **103** when the objects to be detected **100** are not present (or not fully grown). FIG. 10 is a schematic diagram schematically illustrating states of the evaporation of the moisture. FIG. 11 is a graph schematically illustrating a relation between the sensor value and time in the detection of the moisture. FIG. 12 is a graph schematically illustrating a relation between the difference data and time in the detection of the moisture. FIG. 13 is an explanatory chart for explaining the sensor values, the difference data, and the detected initial moisture region in the detection of the moisture.

[0080] As illustrated in FIG. 10, at time T₀ in the initial state, the transmitted light intensity in the area overlapping the moisture **103** on the container **110** is higher than the transmitted light intensity in an area where the moisture **103** is not present (that is, the culture medium **102**). This is due to the following reason. The refractive index n of air is substantially 1. The refractive index n of the container **110** of acrylic is substantially 1.49. The refractive index n of water is substantially 1.33. In an optical path without water (moisture) in the present disclosure, light passes through air (n=1), the container **110** (n=1.49), and then air (n=1) in this order, for example. By contrast, in an optical path through water (moisture), light passes through air (n=1), the container **110** (n=1.49), water (n=1.33), and then air (n=1) in this order, for example. Thus, according to the Fresnel reflectance, the optical path through water (moisture) exhibits higher transmittance than that of the optical path without water (moisture). Herein, the area indicating the presence of the moisture in the initial state (at time T₀) is referred to as the initial moisture region R1.

[0081] At times T₁ and T_i when the predetermined times have elapsed from time T₀, the moisture **103** gradually evaporates and the area of the moisture **103** becomes smaller. At time T₁, the transmitted light intensity in the area where the moisture **103** has evaporated has decreased and becomes equal to the transmitted light intensity of the culture medium **102**. Herein, the area where the moisture **103** has evaporated is referred to as the moisture evaporation region R2. The moisture evaporation region R2 at time T₁ is an area between the outer periphery of the moisture **103** and the outer periphery of the initial moisture region R1. As the area of the moisture **103** decreases, the moisture evaporation region R2 becomes larger.

[0082] At time T_i, when all the moisture **103** evaporates, the entire initial moisture region R1 becomes the moisture evaporation region R2, and the transmitted light intensity of the initial

moisture region **R1** becomes equal to the transmitted light intensity of the culture medium **102**.

[0083] FIG. **11** illustrates the relation between sensor value So and time in the area overlapping the moisture **103** in FIG. **10**. As illustrated in FIG. **11**, the sensor value So at time $T0$ exhibits the highest value. As the moisture **103** evaporates over time, the sensor value So becomes smaller as the transmitted light intensity decreases.

[0084] As illustrated in FIG. **12**, the difference data $Diff$ exhibits a tendency of change inverted from that of the sensor value So illustrated in FIG. **11**. That is, the difference data $Diff$ increases as the moisture **103** evaporates from time $T0$, and after a predetermined period of time has elapsed and the moisture **103** has evaporated, the difference data $Diff$ exhibits a substantially constant value.

[0085] As illustrated in FIG. **13**, the difference data $Diff$ in the area overlapping the moisture **103** at time Ti is zero (illustrated in black). At time Ti , the area that overlaps the initial moisture region **R1** and does not overlap the moisture **103** is the moisture evaporation region **R2**. The difference data $Diff$ in the moisture evaporation region **R2** is larger than that in the moisture **103**. The difference data $Diff$ in an area not overlapping the initial moisture region **R1** (area outside the initial moisture region **R1**) is zero (illustrated in black).

[0086] More specifically, the determination circuit **76** (refer to FIG. **5**) compares the difference data $Diff$ with the first threshold Th_c and the second threshold Th_w stored in advance in the storage circuit **78**. The second threshold Th_w is set to a value smaller than the first threshold Th_c , and is a reference value indicating that the moisture **103** has been detected.

[0087] The determination circuit **76** extracts a first region ($Th_w < Diff$) where the difference data $Diff$ is larger than the second threshold Th_w in a distribution of the difference data $Diff$. The determination circuit **76** determines the first region to be the moisture evaporation region **R2**. Then, when an area where the difference data $Diff$ is smaller than the second threshold Th_w ($Diff < Th_w$) is present in an area (including the moisture evaporation region **R2** and the moisture **103**) inside the outer periphery of the first region, the determination circuit **76** determines the area ($Diff < Th_w$) to be the moisture **103**. When the moisture **103** is detected in the first region ($Th_w < Diff$), the determination circuit **76** determines an area surrounded by the outer periphery of the first region to be the initial moisture region **R1**.

[0088] As described above, the determination circuit **76** detects the presence of the moisture **103** ($Diff < Th_w$) within an area where the difference data $Diff$ exceeds the second threshold Th_w and does not exceed the first threshold Th_c ($Th_w < Diff < Th_c$).

[0089] As illustrated in FIGS. **6** to **13**, the time-varying changes in the sensor value So and the difference data $Diff$ due to the growth of the object to be detected **100** exhibit the same tendency of change as the time-varying changes in the sensor value So and the difference data $Diff$ due to the evaporation of the moisture **103**. Even in this case, the detection device **1** of the present embodiment can determine whether the time-varying changes are due to the moisture **103** or the objects to be detected **100** by comparing the difference data $Diff$ with the first threshold Th_c and the second threshold Th_w in the distribution of the difference data $Diff$.

[0090] FIG. **14** is a graph schematically illustrating a relation between the sensor value and time in the detection of the moisture according to a modification of the embodiment. FIG. **15** is a graph schematically illustrating a relation between the difference data and time in the detection of the moisture according to the modification. FIG. **16** is an explanatory chart for explaining the sensor values, the difference data, and the detected initial moisture region in the detection of the moisture according to the modification.

[0091] FIGS. **14** to **16** illustrate examples of a case where the transmitted light intensity in some region **103a** of the moisture **103** is higher at time $T0$ in the initial state. As illustrated in FIGS. **14** and **15**, the change in the sensor value So due to the evaporation of the moisture may increase in the region **103a**, causing the difference data $Diff$ to exceed the first threshold Th_c .

[0092] As illustrated in FIG. **16**, the determination circuit **76** detects the region **103a** where the

difference data Diff exceeds the first threshold Th_c , in the initial moisture region R1 where the difference data Diff is larger than the second threshold Th_w ($Th_w < Diff < Th_c$). In this case, it is difficult to determine, by using only the difference data Diff, whether the region **103a** is a portion of the moisture **103** or the object to be detected **100**. The alert output circuit **77** outputs the alert indicating the detection of the objects to be detected **100** in the initial moisture region R1, based on the determination result from the determination circuit **76**. This alert causes the user to view the inside of the container **110** to check that not the objects to be detected **100**, but the portion of the moisture **103** has been detected.

[0093] The following describes the method for detecting the objects to be detected **100** when located so as to overlap the moisture **103**. FIG. **17** is a graph schematically illustrating a relation between the sensor value and time in the detection of the objects to be detected and the moisture. FIG. **18** is a graph schematically illustrating a relation between the difference data and time in the detection of the objects to be detected and the moisture. FIG. **19** is a schematic diagram schematically illustrating the states of the growth of the objects to be detected and the states of the evaporation of the moisture. FIG. **20** is an explanatory chart for explaining the sensor values, the difference data, and the objects to be detected and the initial moisture region that have been detected in the detection of the objects to be detected and the moisture.

[0094] The following describes two examples in each of which the objects to be detected **100** overlap the moisture **103**: an example where the objects to be detected **100** grow in the area that overlaps the initial moisture region R1 after the moisture **103** has evaporated; and an example where the objects to be detected **100** grow faster than the evaporation of the moisture **103** (the objects to be detected **100** and the moisture **103** are simultaneously present).

[0095] FIGS. **17** and **18** illustrate the example where the objects to be detected **100** grow after the moisture **103** has the evaporated. As illustrated in FIG. **17**, the sensor value So exhibits the highest value at time $T0$. As the moisture **103** evaporates over time, the sensor value So becomes smaller as the transmitted light intensity decreases. As the objects to be detected **100** grow after the moisture **103** has evaporated, the sensor value So becomes smaller as the transmitted light intensity further decreases.

[0096] As illustrated in FIG. **18**, the evaporation of the moisture **103** makes the difference data Diff larger than the second threshold Th_w and smaller than the first threshold Th_c . After the moisture **103** has evaporated, the growth of the objects to be detected **100** makes the difference data Diff larger than the first threshold Th_c .

[0097] Thus, when the objects to be detected **100** grow after the evaporation of the moisture **103**, the sensor value So and the difference data Diff exhibit two-step changes. In the changes of the sensor value So and the difference data Diff due to the evaporation of the moisture **103**, the same method as that described above with reference to FIGS. **10** to **13** can be applied to determine the initial moisture region R1 and the moisture evaporation region R2. In the changes of the sensor value So and the difference data Diff due to the growth of the objects to be detected **100**, the same method as that described above with reference to FIGS. **6** to **9** can be applied to determine the objects to be detected **100**.

[0098] The determination circuit **76** compares the initial moisture region R1 detected by the evaporation of the moisture **103** with the position of the objects to be detected **100** detected by the growth of the objects to be detected **100**. The determination circuit **76** determines that the objects to be detected **100** are present at a position overlapping the initial moisture region R1. The alert output circuit **77** outputs the alert indicating the detection of the objects to be detected **100** in the initial moisture region R1.

[0099] FIGS. **19** and **20** explain an example where the objects to be detected **100** grow faster than the evaporation of the moisture **103**. As illustrated in FIG. **19**, the moisture **103** gradually evaporates over time from time $T0$. The objects to be detected **100** gradually grow over time from time $T0$ in the area overlapping the initial moisture region R1. At time Ti before the evaporation of

the moisture **103**, the objects to be detected **100** and the moisture **103** are simultaneously present, and the objects to be detected **100** overlap the moisture **103**. At time T_j after a predetermined period of time has elapsed from time T_i , the moisture **103** has evaporated and the objects to be detected **100** are present in the initial moisture region **R1** (moisture evaporation region **R2**). The following description describes the case where the objects to be detected **100** and the moisture **103** are simultaneously present at time T_i .

[0100] As illustrated in FIG. **20**, the determination circuit **76** compares the difference data $Diff$ with the first threshold Th_c and the second threshold Th_w . That is, the determination circuit **76** extracts a region in the distribution of the difference data $Diff$ where the difference data $Diff$ is larger than the first threshold Th_c (hereinafter, referred to as “white region”). If the white region ($Th_c < Diff$) has an area equal to or larger than a predetermined area, the determination circuit **76** determines that the white region ($Th_c < Diff$) is an object-to-be-detected region indicating the detection of the objects to be detected **100**.

[0101] Furthermore, in the same way as the example described with reference to FIG. **13**, the determination circuit **76** determines the initial moisture region **R1**, the moisture evaporation region **R2**, and the moisture **103** by comparing the difference data $Diff$ with the first threshold Th_c and the second threshold Th_w . The method for determining the initial moisture region **R1**, the moisture evaporation region **R2**, and the moisture **103** is the same as that described with reference to FIG. **13**, and will not be repeatedly described.

[0102] The determination circuit **76** compares the initial moisture region **R1** with the position of the objects to be detected **100** (white region ($Th_c < Diff$)) and determines that the objects to be detected **100** overlap the initial moisture region **R1**. The alert output circuit **77** outputs the alert indicating the detection of the objects to be detected **100** in the initial moisture region **R1**. This alert allows the user to visually check the inside of the container **110** based on the alert to check whether the detection is detection of the objects to be detected **100** or false detection of the moisture **103**.

[0103] The following describes a detailed method for detecting the objects to be detected **100** and the moisture **103** with reference to FIGS. **21** to **24**. FIG. **21** is a flowchart for explaining a method for obtaining the baseline data. As illustrated in FIG. **21**, in the initial state after power-on, the control circuit **70** (light source control circuit **72**) turns on the light source **80** (Step **ST11**).

[0104] The control circuit **70** (sensor control circuit **71**) scans the photodetection elements **30** to perform the detection of the optical sensor **10** (Step **ST12**). The photodetection elements **30** output the sensor values So each of which corresponds to the light that has been emitted from the light source **80** and transmitted through the container **110** (at least one item of the objects to be detected **100**, the culture medium **102**, and the moisture **103**).

[0105] The sensor control circuit **71** acquires the sensor values So in the initial state from the photodetection elements **30** and stores the detected sensor values So as the baseline data in the storage circuit **78** (Step **ST13**). FIG. **22** is a flowchart for explaining a method for obtaining the difference data. As illustrated in FIG. **22**, after a predetermined period of time has elapsed from the initial state (for example, at time T_i), the control circuit **70** (light source control circuit **72**) turns on the light source **80** (Step **ST21**).

[0106] The control circuit **70** (sensor control circuit **71**) scans the photodetection elements **30** to perform the detection of the optical sensor **10** (Step **ST22**). The photodetection elements **30** output the sensor values So each of which corresponds to the light transmitted through at least one item of the objects to be detected **100**, the culture medium **102**, and the moisture **103** at time T_i .

[0107] The arithmetic circuit **74** calculates the difference data $Diff$ based on the sensor values So received from the optical sensor **10** and the baseline data received from the storage circuit **78** (Step **ST23**). The calculation of the difference data $Diff$ is performed for each sensor pixel **PX**.

[0108] FIG. **23** is a flowchart for explaining a method for detecting the objects to be detected and the output of the alerts. As illustrated in FIG. **23**, the arithmetic circuit **74** calculates the difference data $Diff$ at a predetermined timing (for example, time T_i) (Step **ST31**). The calculation of the

difference data Diff at Step ST31 is the same as in the flowchart illustrated in FIG. 22. However, FIG. 23 does not illustrate Steps ST21 and ST22 in the flowchart of FIG. 22.

[0109] The image generation circuit 75 generates a binarized image based on the difference data Diff (Step ST32). The image generation circuit 75 compares the difference data Diff with the first threshold Th_c . The image generation circuit 75 sets the sensor pixels PX where the difference data Diff is equal to or larger than the first threshold Th_c ($Th_c \leq Diff$) to be the white region (object-to-be-detected region) and sets the sensor pixels PX where the difference data Diff is lower the first threshold Th_c ($Diff < Th_c$) to be a black region (refer to FIG. 9).

[0110] The determination circuit 76 receives the binarized image from the image generation circuit 75 and determines whether the group of the sensor pixels PX in the white region ($Th_c \leq Diff$) is $N \times N$ pixels or more (Step ST33). N is the number of pixels indicating that the objects to be detected 100 have grown to be a colony and is set based on past detection data of colonies (objects to be detected 100). The number of pixels N is stored in advance in the storage circuit 78.

[0111] If the white region has less than $N \times N$ pixels (No at Step ST33), the determination circuit 76 determines that the objects to be detected 100 have not grown, and ends the process of detecting the objects to be detected 100. In this case, the alert output circuit 77 does not output the alert indicating the detection of the colony (objects to be detected 100).

[0112] If the white region has $N \times N$ pixels or more (Yes at Step ST33), the determination circuit 76 determines that the objects to be detected 100 have grown as the colony and performs processing to detect the initial moisture region R1 (Step ST34).

[0113] FIG. 24 is a flowchart for explaining a method for detecting the moisture. As illustrated in FIG. 24, the determination circuit 76 compares the difference data Diff obtained at Step ST31 with the second threshold Th_w , and extracts the region where the difference data Diff is larger than the second threshold Th_w (first region: $Th_w < Diff$) and a region where the difference data Diff is equal to or smaller than the second threshold Th_w ($Diff \leq Th_w$) (Step ST41).

[0114] The determination circuit 76 determines whether the region where the difference data Diff is larger than the second threshold Th_w (first region ($Th_w < Diff$)) includes another region where the difference data Diff is larger than the second threshold Th_w ($Th_w < Diff$) (Step ST42).

[0115] If the other region ($Th_w < Diff$) is present inside the outermost first region ($Th_w < Diff$) (Yes at Step ST42), the determination circuit 76 sets the outermost first region ($Th_w < Diff$) to be a region A (Step ST43). At Step ST43, the case where the other region ($Th_w < Diff$) is present inside the outermost first region ($Th_w < Diff$) refers to that a ring-shaped bright-to-dark-to-bright pattern is repeatedly arranged from the outer periphery toward the inner periphery in the distribution of the difference data Diff (refer to FIG. 20).

[0116] If the other region ($Th_w < Diff$) is not present inside the outermost first region ($Th_w < Diff$) (No at Step ST42), the determination circuit 76 sets the region ($Th_w < Diff$) as region A (Step ST44). At Step ST44, the case where the other region ($Th_w < Diff$) is not present inside the outermost first region ($Th_w < Diff$) refers to that a ring-shaped bright-to-dark pattern is arranged from the outer periphery toward the inner periphery in the distribution of the difference data Diff (refer to FIG.

[0117] 13), or a case where the region indicating darkness ($Diff \leq Th_w$) is not present inside the first region ($Th_w < Diff$).

[0118] Then, the determination circuit 76 determines whether a second region ($Diff \leq Th_w$) where the difference data Diff is equal to or smaller than the second threshold Th_w is present in region A (Step ST45). If the second region ($Diff \leq Th_w$) is present in region A (Yes at Step ST45), the determination circuit 76 determines region A (that is, the outermost first region ($Th_w < Diff$)) to be the initial moisture region R1 (Step ST46). The determination circuit 76 determines the second region ($Diff \leq Th_w$) in region A to be an area where the moisture 103 has been detected, and determines an area between the outer periphery of the second region ($Diff \leq Th_w$) and the outer periphery of region A to be the moisture evaporation region R2.

[0119] If the second region ($\text{Diff} \leq \text{Th}_w$) is not present in region A (No at Step ST45), the moisture **103** is determined to be not present, and the detection of the moisture ends. In this case, the determination circuit **76** determines region A to be the moisture evaporation region R2 (=initial moisture region R1).

[0120] Referring back to FIG. 23, the determination circuit **76** determines whether the white region (object-to-be-detected region) overlaps the initial moisture region R1 (Step ST35). More specifically, the determination circuit **76** compares the position of the sensor pixel PX indicating the white region obtained at Step ST32 with the initial moisture region R1 obtained at Step ST46.

[0121] If the white region (object-to-be-detected region) overlaps the initial moisture region R1 (Yes at step ST35), the determination circuit **76** outputs the determination result to the alert output circuit **77**. The alert output circuit **77** outputs the alert indicating the detection of the colony (objects to be detected **100**) in the initial moisture region R1, based on the determination result from the determination circuit **76** (Step ST36).

[0122] If the white region (objects to be detected **100**) does not overlap the initial moisture region R1 (No at Step ST35), the determination circuit **76** outputs the determination result to the alert output circuit **77**. That is, the determination circuit **76** determines that the objects to be detected **100** have been detected outside the initial moisture region R1. The alert output circuit **77** outputs the alert indicating the detection of the colony (objects to be detected **100**), based on the determination result from the determination circuit **76** (Step ST37).

[0123] The display device **86** of the external circuit **85** displays the alert and also displays an image indicating the objects to be detected **100** and the initial moisture region R1 or the image indicating the detected location of the objects to be detected **100** (Step ST38).

[0124] As described above, the detection device **1** of the present embodiment compares the difference data Diff with the first threshold Th_c and the second threshold Th_w in the distribution of the difference data Diff to determine, based on the bright/dark pattern of the difference data Diff, whether the objects to be detected **100** have been detected or the objects to be detected **100** in the initial moisture region R1 have been detected.

[0125] The processes illustrated in FIGS. 22 and 23 are repeated at predetermined intervals of time and performed a plurality of times from time T0 in the initial state. This operation allows the detection device **1** to observe changes over time concerning the growth of the objects to be detected **100** and the evaporation of the moisture **103**. Each of the processes described above can be modified as appropriate. For example, in the process of detecting the objects to be detected **100** illustrated in FIG. 23, the detection of the initial moisture region R1 illustrated at Step ST34 may be performed in advance before the detection process in FIG. 23.

[0126] While the preferred embodiment of the present disclosure has been described above, the present disclosure is not limited to the embodiment described above. The content disclosed in the embodiment is merely an example, and can be variously modified within the scope not departing from the gist of the present disclosure. Any modifications appropriately made within the scope not departing from the gist of the present disclosure also naturally belong to the technical scope of the present disclosure. At least one of various omissions, substitutions, and changes of the components can be made without departing from the gist of the embodiment and the modification described above.

Claims

1. A detection device comprising: an optical sensor comprising a plurality of photodetection elements arranged in a planar configuration; a light source configured to emit light to the photodetection elements; and an object placement member that has a light-transmitting property and on which a plurality of objects to be detected are to be placed, wherein the light source, the object placement member, and the optical sensor are arranged in the order as listed, the optical

sensor is configured to obtain baseline data in an initial state after power-on, the detection device is configured to: obtain difference data based on a difference between the baseline data and sensor values obtained from the photodetection elements at intervals of a predetermined period of time; and output, when the difference data has exceeded a first threshold, either an alert indicating detection of the objects to be detected in an initial moisture region indicating presence of moisture in the initial state, or an alert indicating detection of the objects to be detected.

2. The detection device according to claim 1, having a second threshold smaller than the first threshold, and configured to: extract a first region where the difference data is larger than the second threshold in a distribution of the difference data, and determine, when a second region where the difference data is smaller than the second threshold is present in an area inside an outer periphery of the first region, the first region to be the initial moisture region.

3. The detection device according to claim 2, configured to determine an area between the initial moisture region and the second region to be a moisture evaporation region indicating that the moisture has evaporated from the initial state until a time when the predetermined period of time has elapsed and at which the difference data is obtained.

4. The detection device according to claim 2, configured to determine the second region to be an area where the moisture has been detected at a time when the predetermined period of time has elapsed and at which the difference data is obtained.

5. The detection device according to claim 1, configured to: extract an object-to-be-detected region where the difference data is larger than the first threshold in a distribution of the difference data, and output, when the object-to-be-detected region overlaps the initial moisture region, the alert indicating the detection of the objects to be detected in the initial moisture region.

6. The detection device according to claim 1, configured to: extract an object-to-be-detected region where the difference data is larger than the first threshold in a distribution of the difference data, and output, when the object-to-be-detected region does not overlap the initial moisture region, the alert indicating the detection of the objects to be detected.

7. The detection device according to claim 1, wherein the photodetection elements are photodiodes.
